

# Code Modularization Analysis

**Generated:** 2026-03-07 18:41:31  
**Project Root:** /Users/newdavid/Documents/PhysicalRisk  
**Threshold:** Files with more than 300 lines  
**File Types Analyzed:** .css, .js, .py

## Executive Summary

**Total Code Files:** 521  
**Total Lines of Code:** 72,454  
**Files Exceeding 300 Lines:** 45  
**Lines in Large Files:** 17,598 (24.3% of total)

**Recommendation:** Consider refactoring the 45 file(s) listed below.

## Files Requiring Modularization (>300 lines)

| Rank | File Path                                          | Type | Lines | Priority |
|------|----------------------------------------------------|------|-------|----------|
| 1    | tests/confest.py                                   | .py  | 681   | Medium   |
| 2    | app.py                                             | .py  | 612   | Medium   |
| 3    | src/visual/interactivity/trading/market/actions.py | .py  | 553   | Low      |
| 4    | src/port/src/property/property.py                  | .py  | 509   | Low      |
| 5    | tests/routes/governance/mrc.py                     | .py  | 496   | Low      |
| 6    | src/port/rand/thames/mortgage_random.py            | .py  | 491   | Low      |
| 7    | src/port/src/storm/storm.py                        | .py  | 477   | Low      |
| 8    | src/port/src/property/propertyts.py                | .py  | 433   | Low      |
| 9    | src/port/src/property/propertyhc.py                | .py  | 433   | Low      |
| 10   | src/port/src/storm/stormts.py                      | .py  | 428   | Low      |
| 11   | src/routes/trading/stress.py                       | .py  | 425   | Low      |
| 12   | src/routes/prs.py                                  | .py  | 422   | Low      |
| 13   | src/visual/interactivity/gauge/gaugehc/__init__.py | .py  | 412   | Low      |
| 14   | src/models/mortgage/mortgage_pricer.py             | .py  | 412   | Low      |
| 15   | src/visual/interactivity/storm/sp_visual.py        | .py  | 408   | Low      |
| 16   | src/reports/property/property_generator.py         | .py  | 399   | Low      |
| 17   | src/port/cdm/gauge.py                              | .py  | 386   | Low      |
| 18   | src/port/cdm/property.py                           | .py  | 377   | Low      |
| 19   | src/models/risk/risk_assessor.py                   | .py  | 372   | Low      |

|    |                                                              |     |     |     |
|----|--------------------------------------------------------------|-----|-----|-----|
| 20 | src/visual/interactivity/gauge/gaugehc/ghc_stress_charts.py  | .py | 368 | Low |
| 21 | tests/models/risk/risk_assessor.py                           | .py | 364 | Low |
| 22 | src/visual/layer/property_layer.py                           | .py | 363 | Low |
| 23 | src/visual/layer/mortgage_layer.py                           | .py | 363 | Low |
| 24 | src/visual/interactivity/governance/models/mg_validation.py  | .py | 363 | Low |
| 25 | src/visual/interactivity/trading/stress/charts.py            | .py | 362 | Low |
| 26 | src/port/cdm/gaugehd.py                                      | .py | 361 | Low |
| 27 | src/visual/popups/property_popup.py                          | .py | 359 | Low |
| 28 | src/reports/gauge/gauge_generator.py                         | .py | 356 | Low |
| 29 | src/models/trading/delta_engine.py                           | .py | 354 | Low |
| 30 | src/visual/interactivity/gauge/gaugehc/ghc_prs_pricer.py     | .py | 347 | Low |
| 31 | src/models/trading/market_state.py                           | .py | 347 | Low |
| 32 | config.py                                                    | .py | 346 | Low |
| 33 | src/loaders/lookups.py                                       | .py | 346 | Low |
| 34 | src/reports/property/property_page_13_risk_analysis.py       | .py | 345 | Low |
| 35 | tests/visual/phase/popups.py                                 | .py | 341 | Low |
| 36 | src/visual/layer/gauge_layer.py                              | .py | 333 | Low |
| 37 | tests/routes/trading/stress_routes.py                        | .py | 332 | Low |
| 38 | src/visual/core/visualizer.py                                | .py | 329 | Low |
| 39 | src/visual/interactivity/storm/stormportfolio.py             | .py | 327 | Low |
| 40 | src/visual/interactivity/governance/models/mg_detail_tabs.py | .py | 323 | Low |
| 41 | src/port/cdm/mortgage.py                                     | .py | 318 | Low |
| 42 | src/port/rand/thames/gauge/gauge_random.py                   | .py | 310 | Low |
| 43 | src/port/rand/thames/stormts_random.py                       | .py | 307 | Low |
| 44 | src/static/js/backend-handler.js                             | .js | 305 | Low |
| 45 | src/models/trading/pnl_engine.py                             | .py | 303 | Low |

## Refactoring Recommendations

**High Priority (>1000 lines):** Immediate refactoring recommended. These files are likely doing too much and will benefit from breaking into smaller modules.

**Medium Priority (600-1000 lines):** Review for modularization opportunities. Look for distinct functional areas that could be separated.

**Low Priority (300-600 lines):** Monitor for growth. Consider refactoring if complexity increases or multiple concerns are mixed.

### General Strategy:

- Identify distinct responsibilities within each file
- Extract cohesive functionality into separate modules
- Create clear interfaces between modules

- Maintain single responsibility principle
- Add tests before and after refactoring

## Complete File Analysis

**.css files:** 2 files, 83 total lines, 42 avg lines/file, 0 file(s) >300 lines

**.js files:** 8 files, 1,109 total lines, 139 avg lines/file, 1 file(s) >300 lines

**.py files:** 511 files, 71,262 total lines, 139 avg lines/file, 44 file(s) >300 lines